

Q. Draw and explain the timing diagram for a Memory Read and Memory Write operation of the 8086 microprocessor in Minimum Mode. Detail the events occurring in each T-State.

📌 Definition to Remember

A **Timing Diagram** graphically represents the signals the CPU generates over time. In 8086, a basic bus cycle consists of 4 clock periods called **T-States (T1, T2, T3, T4)**. T1 is for addressing, T2 for bus turnaround, T3 for data transfer, and T4 for cycle completion.

1. General T-State Roles

- **T1:** Address is generated and latched.
- **T2:** Multiplexed bus switches from Address to Data mode. Control signals asserted.
- **T3:** Actual data transfer occurs. (Wait states T_W inserted here if memory is slow).
- **T4:** Control signals deactivated; cycle ends.

2. Memory Read Timing (Minimum Mode)

CPU fetches data from memory.

* **T1:**

* 20-bit address placed on AD and A bus.

* $\overline{M}/\overline{IO}$ is **HIGH** (Memory op).

* $\overline{ALE}$ (Address Latch Enable) pulses **HIGH** to latch the address.

* $\overline{DT}/\overline{R}$ goes **LOW** (Configure transceivers to Receive).

* **T2:**

* Address is removed from AD bus (goes to high impedance).

* $\overline{RD}$ (Read) is pulled **LOW**.

* $\overline{DEN}$ (Data Enable) goes **LOW** to enable transceivers.

* **T3:**

* Memory places data onto the AD bus. CPU reads it.

* (CPU checks *READY* pin. If low, inserts T_W).

* **T4:**

* $\overline{RD}$ and $\overline{DEN}$ go **HIGH**. Data is locked into the CPU. Cycle ends.

3. Memory Write Timing (Minimum Mode)

CPU sends data to memory.

* **T1:**

* 20-bit address placed on AD and A bus.

* $\overline{M}/\overline{IO}$ is **HIGH**. $\overline{ALE}$ pulses **HIGH**.

* $\overline{DT}/\overline{R}$ goes **HIGH** (Configure transceivers to Transmit).

* **T2:**

* CPU places data to be written onto the AD bus.

* $\overline{WR}$ (Write) is pulled **LOW**.

* $\overline{DEN}$ is pulled **LOW**.

* **T3:**

* Data remains stable on the AD bus so memory has time to write it.

* **T4:**

* $\overline{WR'}$ and $\overline{DEN'}$ go **HIGH**. AD bus goes to high impedance. Cycle ends.

★ Must-Write Points (for 10 marks)

1. A basic bus cycle consists of 4 clock periods: **T1, T2, T3, T4**.
2. **T1**: CPU outputs address on AD bus; pulses **ALE** high to latch the address.
3. **T2**: AD bus switches to data. Control signals ($\overline{RD'}$ or $\overline{WR'}$, and $\overline{DEN'}$) are asserted (pulled Low).
4. **T3**: Data is transferred (Read from or written to memory).
5. **Wait States (T_W)**: Inserted between T3 and T4 if the **READY** pin is low (used for slow memory).
6. **T4**: Control signals are deactivated (pulled High) and the cycle concludes.
7. **DT/R'** indicates direction (Low for Read, High for Write). $\overline{M/IO'}$ is High for Memory.

⚡ Quick Recall

Timing Diagram → 4 T-States → T1 (Address out, ALE pulse) → T2 ($\overline{RD/WR}$ asserted, $\overline{DEN}$ asserted, Bus turnaround) → T3 (Data Transfer, Wait States if slow) → T4 (Signals deactivated)